SQUID Characterisation and Shapiro Resonance

Laboratory Notes

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1. Single Junction V-I Characteristics (Tasks 1 and 2)

Experimental Setup The sample was cooled into the superconducting (SC) regime. Measurements were driven by a 120 µA triangular wave bias current through a 10 kΩ bias resistor, producing a linear current sweep.

Results The transition mapped the energy gap $\Delta(T) \propto T_c\sqrt{1 - T/T_c}$, indicating the binding energy of Cooper pairs. Parameter extraction from the threshold yielded a critical current $I_c = 82\,\mu\text{A}$ (see Appendix A, Fig. 1).

2. SQUID Flux Response and Inductance (Task 3)

Setup A 41 Hz triangle drive modulated the flux bias.

Results The interference response occurred at approximately 400(100) Hz, signalling traversal across multiple flux quanta within a single sweep (see Appendix A, Fig. 2).

Mutual Inductance Extraction Automated peak detection on 14 analyzed periods yielded bias periods $\Delta V_{\text{ref}} = 0.8293\,\text{V}$ and $\Delta I_{\text{flux}} = 82.93\,\mu\text{A}$. Using $\Delta I_{\text{flux}} = \Phi_0/M$, the extracted mutual inductance is $M = 24.94\,\text{pH}$, consistent with manual estimations.

3. Shapiro Resonances: Frequency Sweep (Tasks 4 and 5)

Objective Verification of the AC Josephson effect relation $\Delta V = \frac{h}{2e}\nu = \Phi_0\nu$.

Setup The SQUID was irradiated with a microwave field; frequency ν was swept from 8 GHz to 12 GHz. Differential amplifier gain ($A \approx 1000\times$) was compensated to retrieve absolute quantum voltage scales.

Data Processing Step size (ΔV) extraction utilized two parallel cross-verifying methodologies: transition edge detection via differential resistance maxima (dV/dI peaks), and plateau centre evaluation via voltage density of states (histogram peaks). While physically grounded, the reliance on fixed peak-prominences and histogram bin-widths introduces ad-hoc heuristics. Implementing adaptive noise-floor thresholding (e.g., $3\sigma_{\text{noise}}$) or Kernel Density Estimation (KDE) would rigorously automate and eliminate bias.

Results and Error Visualisation of the 10 GHz step extraction workflow is given in Appendix A, Fig. 3. The frequency dependence yielded an extracted $\Phi_0 = 1.88 \times 10^{-15}\,\text{Wb}$ (Error: $\sim$ 9.2%). However, despite this seemingly low aggregate error, the underlying ΔV vs. ν data exhibits severe scatter (see Appendix A, Fig. 4). The ad-hoc nature of the fixed-prominence edge detection fails to adapt to varying noise floors and impedance mismatches across the frequency sweep, resulting in catastrophic misidentification of plateau boundaries. Consequently, the extracted voltage step sizes are highly unstable and erratic, demonstrating the fragility of rigid heuristic processing on raw cryogenic signals.

4. Shapiro Resonances: Power Sweep (Task 6)

Setup Microwave frequency fixed at 10 GHz whilst iterating driving power levels. 28 measured step points were mapped onto 8 discrete power tiers (-26.1 dBm to 11.7 dBm).

Processing Microwave driver power P was systematically interpolated across the measurement index N using a robust 3rd-order polynomial map $P(N) = a_3N^3 + a_2N^2 + a_1N + a_0$. Extracted ΔV magnitudes from histogram/derivative methods were mapped sequentially against the interpolated indices.

Results The cubic polynomial ($a_3 \approx 3.78 \times 10^{-3}$, $a_2 \approx -0.20$, $a_1 \approx 3.90$, $a_0 \approx -25.22$) generated a reliable stability curve (RSS ≈ 7.53) verifying standardisation without exhibiting Runge's phenomenon or high-frequency over-oscillation (see Appendix A, Fig. 5).

A. Figures

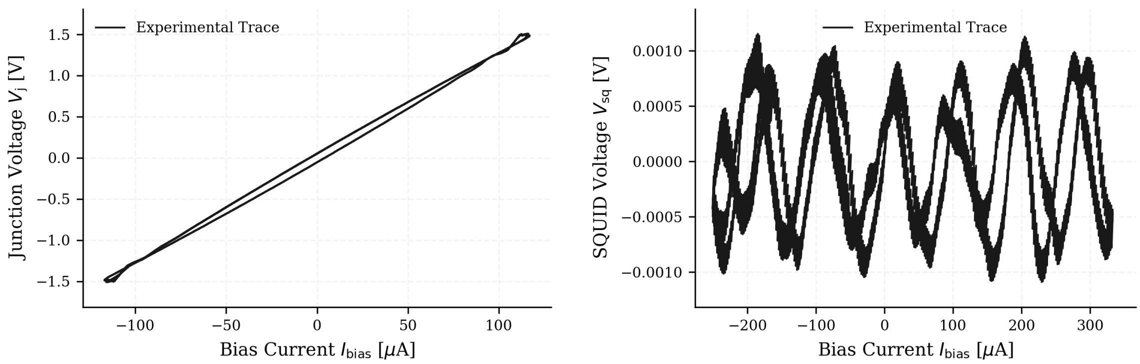

Figure 1: Left: Single Josephson Junction I-V trace mapping the SC density of states. Right: SQUID I-V response under magnetic flux tuning ($I_b = \pm 300\,\mu$A). Adaptive gain applied.

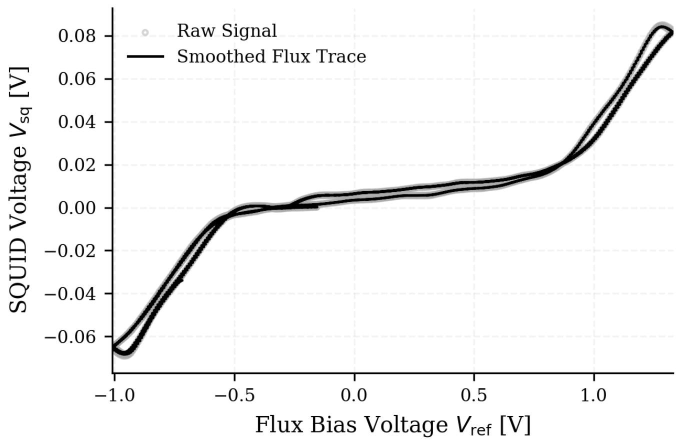

Figure 2: Flux interference pattern demonstrating macroscopic quantum coherence.

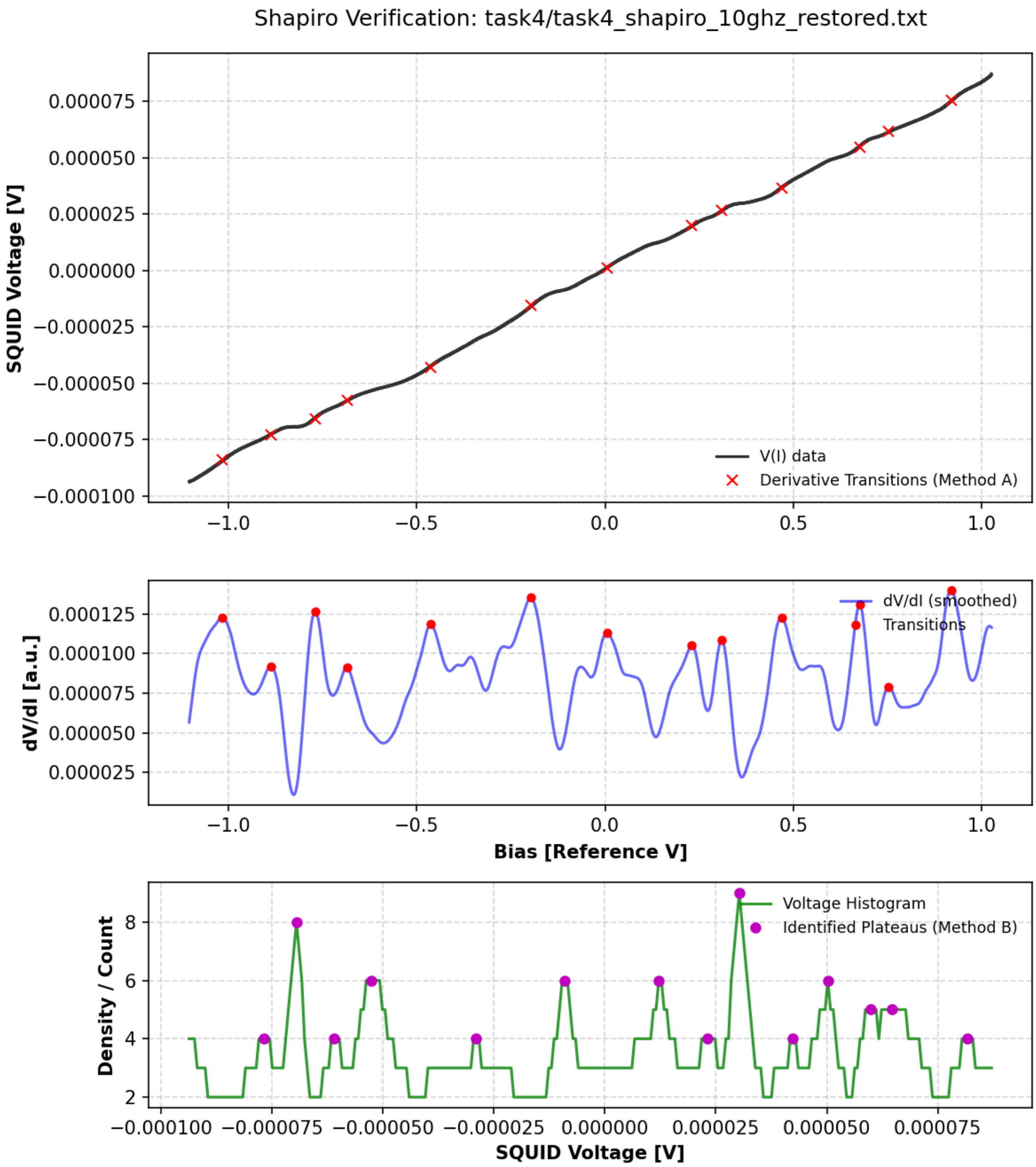

Figure 3: Shapiro step verification at 10 GHz. Panels display the raw sweep (top), parametric differential resistance bounding (middle), and plateau density of states extraction (bottom).

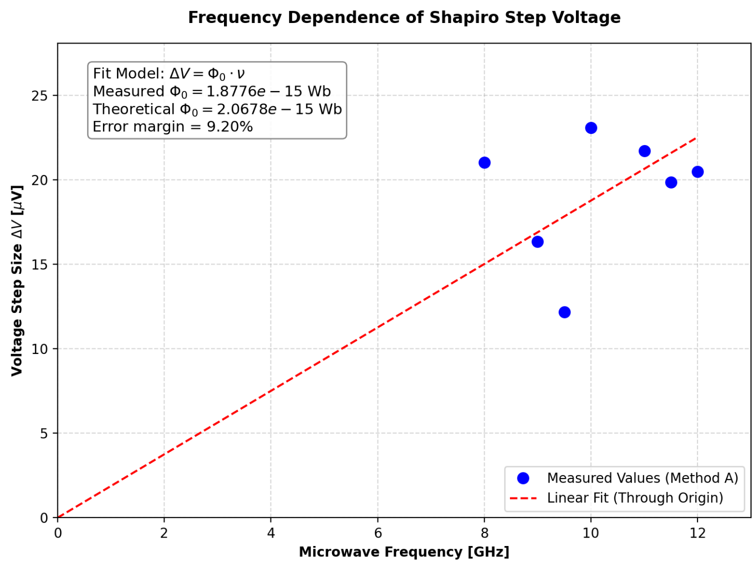

Figure 4: Linear dependence of the fundamental Shapiro voltage step magnitude on applied microwave frequency, fitted through the origin.

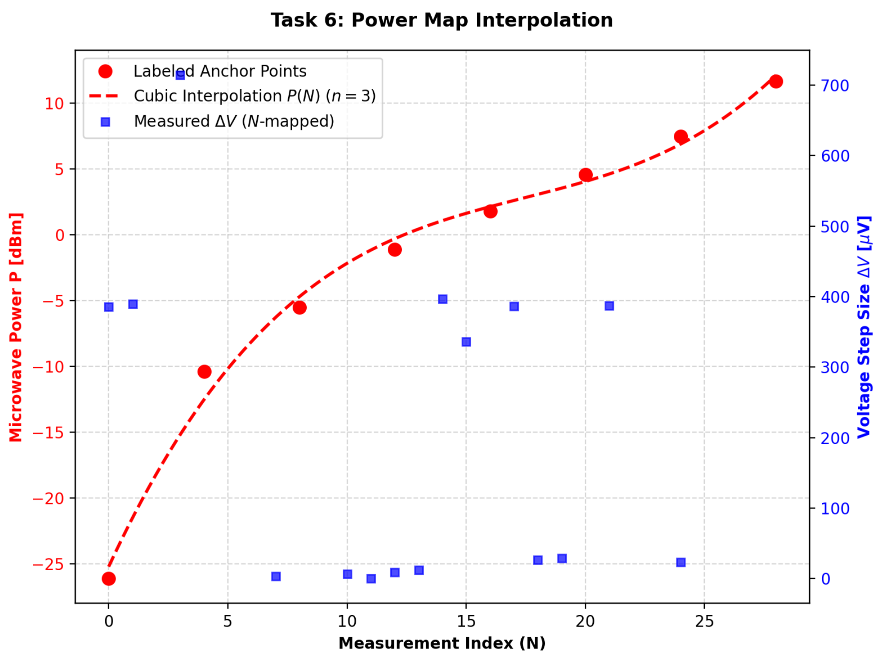

Figure 5: Interpolated power mapping (P vs N) utilising a resilient 3rd-order fit, alongside simultaneously extracted 10 GHz fundamental Shapiro step magnitudes (ΔV).